TWO-ENDS GEOMETRY AND MULTILINEAR METHODS IN FOURIER RESTRICTION

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Abstract. We develop a streamlined framework for proving restriction estimates, which yields stronger results than those obtained by polynomial partitioning methods in higher dimensions.

1. Introduction

Let B^{n-1} be the unit ball in $\mathbb{R}^{n-1}$, and define

$$Ef(x', x_n) = \int_{B^{n-1}} e^{i(x'\cdot\xi + x_n|\xi|^2)} f(\xi)\, d\xi.$$

The diagonal restriction problem asks for the range of p for which

$$\|Ef\|_{L^p(\mathbb{R}^n)} \leqslant C_{n,p}\|f\|_{L^p(B^{n-1})}. \tag{1.1}$$

The conjectured range is $p > 2n/(n-1)$. Throughout, $n \geqslant 5$ denotes the ambient dimension. We consider only the constant-coefficient elliptic paraboloid, not the cone or variable-coefficient operators.

Polynomial partitioning has led to a sequence of advances in this problem; see, for example, [6, 8, 5]. Wang–Wu [12] introduced a different approach based on two-ends incidence geometry and refined decoupling. We combine this approach with the broad–narrow method [2, 6], inserting Zahl's multilinear Kakeya estimate [13] directly into the broad argument. This avoids the loss that can occur when the multilinear estimate is first converted into a linear Kakeya maximal estimate.

The construction is inspired by the organization of incidences in planebrush arguments [9, 1, 10]. Instead of establishing an unrestricted higher-dimensional planebrush estimate, we work with transverse tube families, retain their possibly unequal populations, and handle angular concentration through narrow decoupling. The argument then passes from a mixed broad estimate to a mixed linear estimate, and finally to (1.1) by interpolation.

Fix

$$2 \leqslant k \leqslant \left\lfloor \frac{n+2}{2} \right\rfloor, \qquad d = d_{n,k} := \frac{n^2 + k^2 + n - k}{2n}, \tag{1.2}$$

and put $D = (4n-5)(d-1) + (n-1)(n-3)$. Define

$$p_b(n,k) = 2 + \frac{4(2d+n-5)}{D}, \tag{1.3}$$

$$p_N(n,k) = 2 + \frac{4}{2n-k}, \tag{1.4}$$

Here p_b is the threshold for the transverse broad estimate, whereas p_N is the threshold required by the narrow induction.

Theorem 1.1. *For $n \geqslant 5$, the estimate* (1.1) *holds whenever*

$$p > \min_{2\leqslant k\leqslant \lfloor (n+2)/2\rfloor} \max\{p_b(n,k), p_N(n,k)\} = 2 + \frac{2.58631445\ldots}{n} + O\left(\frac{1}{n^2}\right).$$

For $n = 9$, our result yields the best-known restriction exponents $p > 2150/929 = 2.314316469\ldots$.

The broad–narrow step is a substitute for the higher-dimensional planebrush input that would be needed for a direct geometric approach. Such an estimate could instead strengthen the Wang–Wu framework beyond the present method.

1.1. **Structure of the article.** Section 2 records the required geometric and analytic tools. Section 3 establishes the multilinear incidence estimate and the transverse analytic bound. Section 4 proves the mixed broad estimate and completes the mixed broad-to-linear induction.

1.2. **Notation and conventions.** We write $A \lesssim B$ when $A \leqslant CB$ with a constant depending only on fixed parameters. At a geometric scale δ, the notation $A \lessapprox B$ allows $C_\epsilon \delta^{-\epsilon}$ for every $\epsilon > 0$, together with any explicitly stated $\delta^{-C\epsilon_1}$ loss from a two-ends hypothesis; the analogous convention is used at physical scale r. Fixed powers of dynamic densities or angles are not absorbed in this notation. We keep logarithmic losses explicit when tracking the broad parameter. A term $\mathrm{RapDec}(r)$ is bounded by $C_N r^{-N}$ for every prescribed N.

A unit δ-tube is a fixed constant enlargement of a unit line segment by transverse radius δ. We represent paraboloid tube directions with positive last coordinate. An m-parallel, direction-separated family has at most m tubes in each δ-separated angular class; bounded direction colorings account for buffered caps.

2. Preliminaries

2.1. **Two-ends shadings.** A shading $Y(T)$ is a union of cells of a fixed δ-grid contained in a fixed enlargement of T. Let

$$E_Y = \bigcup_{T\in\mathbb{T}} Y(T), \qquad \mu_Y(x) = \sum_{T\in\mathbb{T}} \mathbb{1}_{Y(T)}(x).$$

The shading is λ-dense if $|Y(T)| \asymp \lambda|T|$ for every retained tube. Later, for a refined subfamily, λ may instead denote the explicitly defined *average* density; this does not impose a new two-ends hypothesis on that subfamily.

Definition 2.1 (Two ends)**.** *For $0 < \epsilon_2 < \epsilon_1 < 1$, a shading is (ϵ_1, ϵ_2)-two-ends if*

$$|Y(T) \cap B(x, \delta^{\epsilon_1})| \leqslant C\delta^{\epsilon_2}|Y(T)|$$

for every T and every such ball. Harmless fixed dilations and fixed constants are incorporated by decreasing ϵ_2 slightly. The constants in the geometric inputs may depend on ϵ_2.

The Wolff axiom with error m means that, for every box of dimensions $1 \times s_1 \times \cdots \times s_{n-1}$, with $\delta \leqslant s_j \leqslant 1$, the number of tubes contained in a fixed dilation of the box is at most

$$Cm\, s_1 \cdots s_{n-1} \delta^{-(n-1)}. \tag{2.1}$$

An m-parallel direction-separated family has this property: a contained tube must point in an angular region having the indicated product size, and each allowable direction contributes at most m tubes. In particular,

$$\delta^{n-1}\#\mathbb{T} \lesssim m.$$

2.2. **Geometric tools.** We use the following two geometric tools with their small-loss versions.

Theorem 2.2 (General two-ends hairbrush; Gao–Wu–Xi)**.** *Suppose that $\mathbb{T}$ satisfies* (2.1) *and has a λ-dense, (ϵ_1,ϵ_2)-two-ends shading. Then* [4, Lemma 4.1] *gives*

$$|E_Y| \gtrsim_\epsilon \delta^{\epsilon+C\epsilon_1} m^{-\frac{n}{2n-2}} \lambda^{7/4}\delta^{(n-2)/2}(\delta^{n-1}\#\mathbb{T})^{\frac{n}{2n-2}}.$$

There is a good set $E_H \subset E_Y$ with $|E_Y\backslash E_H| \leqslant \delta^{\epsilon_1}|E_Y|$ on which

$$\mu_Y \lesssim_\epsilon \delta^{-\epsilon-C\epsilon_1} m^{\frac{n}{2n-2}} \lambda^{-3/4}\delta^{-(n-2)/2}(\delta^{n-1}\#\mathbb{T})^{\frac{n-2}{2n-2}}.$$

Consequently, for an m-parallel direction-separated family,

$$\mu_Y \lessapprox m\lambda^{-3/4}\delta^{-(n-2)/2} \quad \text{on } E_H. \tag{2.2}$$

Only this general form of the hairbrush estimate is used below.

Theorem 2.3 (Direction-separated multilinear Kakeya; Zahl)**.** *Let $2 \leqslant k \leqslant n$ and $d = d_{n,k}$. For a direction-separated family of unit δ-tubes,* [13, Theorem 1.3] *gives*

$$\left\| \left(\sum_{T_1,\dots,T_k\in\mathbb{T}} \mathbb{1}_{T_1}\cdots\mathbb{1}_{T_k}|v(T_1)\wedge\cdots\wedge v(T_k)|^{k/d} \right)^{1/k} \right\|_{d/(d-1)} \lesssim_\epsilon \delta^{1-n/d-\epsilon}(\delta^{n-1}\#\mathbb{T})^{\frac{n(d-1)}{(n-1)d}}. \tag{2.3}$$

We use (2.3) directly, without invoking the multilinear-to-linear Kakeya reduction.

2.3. **Broad norms.** Fix $R \gg 1$ and a dyadic $K \asymp R^\kappa$, with $\kappa > 0$ small. Let τ range over a finitely overlapping cover of B^{n-1} by K^{-1}-caps. We write $\boldsymbol{f} = (f_\tau)_\tau$, where $\operatorname{supp} f_\tau \subset 2\tau$, and use the notation

$$\|\boldsymbol{f}\|_2^2 = \sum_\tau \|f_\tau\|_2^2, \qquad \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}^2 = \frac{1}{|\theta|}\sum_\tau \int_\theta |f_\tau|^2. \tag{2.4}$$

For a smooth cap decomposition of f, these quantities are bounded by the corresponding quantities for f, up to a fixed constant.

For a cube b of side K^2, define

$$\mu_{k,A,p}(E\boldsymbol{f};b) = \inf_{V_1,\dots,V_A} \max_{\substack{\tau:\,\measuredangle(v(2\tau),V_a)>K^{-1}\\ \text{for all } 1\leqslant a\leqslant A}} \int_b |Ef_\tau|^p, \tag{2.5}$$

where the infimum is over $(k-1)$-dimensional linear subspaces of $\mathbb{R}^n$, and $v(2\tau)$ denotes the tube directions associated with 2τ. The maximum is zero if there is no eligible cap. Set

$$\|E\boldsymbol{f}\|^p_{\mathrm{BL}^p_{k,A,K}(U)} = \sum_{b\subset U} \mu_{k,A,p}(E\boldsymbol{f};b).$$

We use a fixed grid and take larger cubes to be unions of its K^2-cubes. The same definition applies when the functions Ef_τ are replaced by other functions indexed by the same caps.

At an intermediate scale r, wave-packet decompositions enlarge frequency supports by $O(r^{-1/2})$. We keep the original cap indices and choose the scales so that these supports remain in 2τ.

The broad norm satisfies the triangle inequality with a loss in the parameter A; see [6, (1.15)].

Lemma 2.4 (Broad triangle inequality)**.** *Let $\boldsymbol{f} = \sum_{j=1}^{L} \boldsymbol{f}_j$ and $A_1 + \cdots + A_L \leqslant A$. Then*

$$\|E\boldsymbol{f}\|_{\mathrm{BL}^p_{k,A,K}(U)} \leqslant \sum_{j=1}^{L} \|E\boldsymbol{f}_j\|_{\mathrm{BL}^p_{k,A_j,K}(U)}.$$

The broad triangle inequality allows us to treat different wave-packet families separately. The following lemma captures the geometric content of broadness by extracting k quantitatively transverse caps.

Lemma 2.5 (Transverse caps)**.** *After subdividing the K^{-1}-caps into smaller caps, the broad contribution is bounded, up to $K^{O(1)}$, by a sum over $K^{O(1)}$ tuples of k caps satisfying*

$$|v_1 \wedge \cdots \wedge v_k| \gtrsim K^{-O(1)} \tag{2.6}$$

for every choice of a direction from each cap. For a fixed tuple, write G_i for the sum over its i-th cap. Its contribution on X is bounded by $K^{O(1)}$ times the left-hand side of

$$\sum_{b \subset X} \min_{1 \leqslant i \leqslant k} \int_b |G_i|^p \leqslant \left(\prod_{i=1}^{k} \int_X |G_i|^p \right)^{1/k}. \tag{2.7}$$

Proof. Subdivide each cap to radius K^{-L}, where $L > 2k + 10$ is fixed. At each b, choose caps successively outside a $(k-1)$-plane containing the directions already chosen. The definition of the broad measure and Minkowski's inequality ensure that each selected cap carries at least $K^{-O(1)}$ times the broad contribution. The successive angular separations imply (2.6); the subdivision ensures that it holds for all directions in the selected caps. Group cubes according to their selected tuple and apply Hölder to obtain (2.7). □

When partitioning X by multiplicity, we first retain the sum on the left of (2.7), which is additive over the cubes, and apply Hölder only after choosing a piece. For wave-packet sums, the smaller caps are formed by grouping packets according to their directions. We require $r^{-1/2} \ll K^{-L}$ so that their frequency supports remain transverse.

2.4. **Wave packets, local averages of the L^2-norm, and refined decoupling.** At scale r, decompose each f_τ into wave packets f_T associated with $r^{-1/2}$-caps θ and tubes T of length r and radius $r^{1/2+\epsilon_0}$. The cap indices are retained throughout. We suppress rapidly decreasing errors and keep the losses $r^{O(\epsilon_0)}$ caused by the tube enlargement.

Almost orthogonality gives, for every packet subfamily $\mathcal{S}$,

$$\left\| \sum_{T \in \mathcal{S}} f_T \right\|_2^2 \lesssim \sum_{T \in \mathcal{S}} \|f_T\|_2^2, \qquad \sum_{T} \|f_T\|_2^2 \lesssim \|\boldsymbol{f}\|_2^2.$$

Together with frequency localization, this gives, for $1 \leqslant t \leqslant r$,

$$\max_{\theta:\, t^{-1/2}\text{-cap}} \Big\| \sum_{T\in\mathcal{S}} f_T \Big\|_{L^2_{\mathrm{avg}}(\theta)} \lesssim \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}. \tag{2.8}$$

These estimates apply cap by cap, with the norms interpreted as in (2.4) when the cap indices are retained.

After dyadic pigeonholing, assume

$$\|f_T\|_2 \asymp a, \qquad \#\{T : \theta(T) = \theta\} \asymp m \tag{2.9}$$

for each occupied cap. Then

$$\begin{aligned} a^2 m &\lesssim r^{-(n-1)/2} \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^2_{L^2_{\mathrm{avg}}(\theta)}, \\ a^2 \#\mathbb{T} &\lesssim \|\boldsymbol{f}\|_2^2. \end{aligned} \tag{2.10}$$

Further subfamilies have at most $O(m)$ packets in each cap and still satisfy (2.10). The pigeonholing costs at most $(\log r)^{O(1)}$, including the loss in A from Lemma 2.4.

Let X be a union of $r^{1/2+\epsilon_0}$-cubes, and let

$$G_i = \sum_{T\in\mathbb{T}_i} Ef_T, \qquad N_i = \#\mathbb{T}_i,$$

where $\mathbb{T}_i$ is a fixed packet subfamily. For each $T \in \mathbb{T}_i$, let $Y_i(T)$ be the union of the cubes of X meeting T, and define

$$\sum_{T\in\mathbb{T}_i} |Y_i(T)| = \lambda_i \sum_{T\in\mathbb{T}_i} |T|.$$

Fixed dilations of cubes and tubes are understood. Local L^2 orthogonality on each cube, summed over X, yields

$$\int_X |G_i|^2 \lesssim r^{O(\epsilon_0)} \lambda_i r\, a^2 N_i. \tag{2.11}$$

Set $q = 2(n+1)/(n-1)$. If every cube of X meets at most μ_i tubes from $\mathbb{T}_i$, refined decoupling [7, Theorem 4.2], in the form [12, Theorem 4.4], gives

$$\begin{aligned} \int_X |G_i|^q &\lesssim_\epsilon r^{\epsilon+O(\epsilon_0)} \mu_i^{2/(n-1)} a^q N_i \\ &\lesssim_\epsilon r^{-1+\epsilon+O(\epsilon_0)} (\mu_i/m)^{2/(n-1)} a^2 N_i \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{q-2}_{L^2_{\mathrm{avg}}(\theta)}. \end{aligned} \tag{2.12}$$

Here $\|Ef_T\|^q_{L^q(w_{B_r})} \lesssim a^q$ follows by interpolating the standard packet bounds $\|Ef_T\|_\infty \lesssim r^{-(n-1)/4} a$ and $\|Ef_T\|^2_{L^2(w_{B_r})} \lesssim r a^2$. The second inequality follows from (2.10). Any dyadic summation over packet L^q-norms is absorbed into r^ϵ.

3. Multilinear two-ends estimates

Fix n, k as in (1.2). Throughout this section,

$$d = d_{n,k}, \qquad q = \frac{2(n+1)}{n-1}, \qquad D = (4n-5)(d-1) + (n-1)(n-3),$$

and p_b, p_N are defined in (1.3)–(1.4). We first estimate the multiplicity of transverse tube families without assuming that their populations are comparable, then combine this estimate with the hairbrush bound and refined decoupling.

3.1. **A geometric lemma allowing unequal transverse populations.**

Lemma 3.1. *Let $\mathbb{T}_1, \dots, \mathbb{T}_k$ be disjoint angular families of unit δ-tubes, and suppose every choice $T_i \in \mathbb{T}_i$ has wedge at least $\nu > 0$. There are at most m tubes in each δ-separated direction class. Let E be a union of $O(\delta^{-n})$ grid cells of side δ. Assume the shaded multiplicity in family i is comparable to a positive constant μ_i on every cell of E. Set*

$$N_i = \#\mathbb{T}_i, \qquad N = \sum_i N_i, \qquad Z = \frac{\delta^{n-1}N}{m},$$

$$\zeta = \frac{(N_1 \cdots N_k)^{1/k}}{N}, \qquad \Lambda = (\lambda_1 \cdots \lambda_k)^{1/k}, \qquad \mu = (\mu_1 \cdots \mu_k)^{1/k}.$$

where $\mu_i|E| \asymp \lambda_i \delta^{n-1} N_i$ defines the average densities. If $\min_i \mu_i/m \geqslant C_{n,k} \log(2/\delta)$, then

$$\frac{\mu}{m} \lesssim_\epsilon \nu^{-1} \delta^{-(n-d)-\epsilon} \Lambda^{-(d-1)} \zeta^{-(d-1)} Z^{(d-1)/(n-1)}. \tag{3.1}$$

Proof. Pad each direction class with dummy entries to have m slots, and choose one slot uniformly and independently in each class. For each fixed cell and family i, the sampled shaded multiplicity is a sum of independent Bernoulli variables, one for each direction class, of total expectation comparable to μ_i/m. Chernoff's inequality and a union bound over the polynomially many cells show that all these sampled multiplicities are at least $c\mu_i/m$ simultaneously, with probability greater than $3/4$, when the displayed logarithmic threshold is large enough. The expected total number of sampled tubes is N/m; Markov's inequality makes it at most $8N/m$ with probability at least $7/8$. Thus a realization with both properties exists. A bounded direction coloring handles adjacent direction cells when necessary.

The chosen realization is direction-separated. Apply (2.3) to its complete tubes, using their shaded incidences only for a lower bound on the integrand on E. Raise the result to $d/(d-1)$:

$$\nu^{1/(d-1)} (\mu/m)^{d/(d-1)} |E| \lesssim_\epsilon \delta^{-(n-d)/(d-1)-\epsilon} Z^{n/(n-1)}. \tag{3.2}$$

Taking geometric means of the incidence identities gives

$$\mu|E| \asymp \Lambda\zeta\delta^{n-1}N = \Lambda\zeta m Z.$$

Substitute this into (3.2), cancel, and raise to the power $d-1$. This proves (3.1), after renaming the arbitrary small loss. □

Remark 3.2. *The lemma does not require comparable N_i or comparable μ_i. It makes no two-ends assumption on the refined shadings. These two features are essential in the application. For enlarged wave-packet tubes, grouping fine directions at the enlarged angular scale replaces m by at most $r^{C_n\epsilon_0}m$; applying the lemma and translating back introduces only $r^{C_{n,k}\epsilon_0}$ factors. The low-multiplicity alternative is treated directly in Subsection 3.2.*

3.2. **The transverse analytic branch.** Assume we have reached a common region X and k transverse packet families as in Subsection 2.4 and Lemma 3.1. We also assume a good-set hairbrush upper bound for the *ancestor shading*:

$$\mu_i \leqslant \mu_{\text{ancestor}} \lesssim r^e m \lambda_0^{-3/4} r^{(n-2)/4}, \tag{3.3}$$

where $e > 0$ is arbitrarily small apart from a fixed dimensional multiple of the two-ends scale parameter. Every refined shading has average density at most $C\lambda_0$. In particular $\Lambda \lesssim \lambda_0$, and (3.3) remains valid with $\lambda_0^{-3/4}$ replaced by $C\Lambda^{-3/4}$.

Here $a^2 N \lesssim \|\boldsymbol{f}\|_2^2$. The numbers Z and ζ are bounded above by dimensional constants and by 1, respectively. The small enlargement errors in Z can instead be included in r^e. We estimate the local minimum functional from Lemma 2.5; an upper bound by a geometric mean of total integrals is invoked only on the selected common region.

If some $\mu_i/m \lesssim r^{C\epsilon_0} \log r$, use this family's L^2 bound with density at most one and its refined decoupling bound. Interpolation with $t = (p_b - 2)/(q - 2) = 1 - u$ gives a factor r^{1-2t+e}. Here

$$u = \frac{(2n-3)(d-1)}{D} < \frac{1}{2},$$

so the exponent without e is strictly negative. The local minimum functional is at most the integral of this one family over the selected region, hence this case satisfies the desired bound. We do not claim to control the geometric mean of all sector integrals by the low-multiplicity family.

In the remaining case apply Lemma 3.1. Put

$$\mathcal{I}_q = \left(\prod_{i=1}^{k} \int_X |G_i|^q \right)^{1/k}.$$

Taking geometric means of (2.11), and then of the two alternatives in (2.12), gives

$$\begin{aligned}
\mathcal{I}_2 &\lesssim r^e \, r \Lambda \zeta a^2 N, \\
\mathcal{I}_q &\lesssim r^e \, r^{-n/(2(n-1))} \Lambda^{-3/(2(n-1))} \zeta a^2 N \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}^{q-2}, \\
\mathcal{I}_q &\lesssim K^C r^e \, r^{-(d-1)/(n-1)} \Lambda^{-2(d-1)/(n-1)} \zeta^{1-2(d-1)/(n-1)} Z^{2(d-1)/(n-1)^2} \\
&\qquad \times a^2 N \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}^{q-2}.
\end{aligned}$$

The polynomial K loss includes the wedge bound of Lemma 2.5.

Choose

$$u = \frac{(2n-3)(d-1)}{D}, \qquad v = \frac{2(n-1)(d-1)}{D}, \qquad w = \frac{(n-1)(n-3)}{D}.$$

These weights are nonnegative, sum to one, and satisfy

$$\begin{aligned}
u - \frac{n}{2(n-1)} v - \frac{d-1}{n-1} w &= 0, \\
u - \frac{3}{2(n-1)} v - \frac{2(d-1)}{n-1} w &= 0.
\end{aligned}$$

Also $2u + q(v + w) = p_b$. Hölder and the two upper bounds for $\mathcal{I}_q$ yield

$$\begin{aligned}
\mathcal{I}_{p_b} &\lesssim K^C r^e \zeta^{\vartheta} Z^{\psi} a^2 N \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}^{p_b-2} \\
&\lesssim K^C r^e \|\boldsymbol{f}\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}^{p_b-2},
\end{aligned} \tag{3.4}$$

where

$$\vartheta = \frac{(2n+1)(d-1) + (n-1)(n-3)}{D} > 0, \qquad \psi = \frac{2(d-1)(n-3)}{(n-1)D} \geqslant 0.$$

This is the precise reason the imbalance in transverse tube populations does not cost a power of the scale. Dropping ζ earlier would not be valid.

4. Proof of the main theorem

4.1. **The broad two-ends induction.** We use the transverse bound of Subsection 3.2 in a two-ends induction for the broad measures (2.5). The outer scale remains fixed while the intermediate radius decreases, and a slowly growing broad parameter pays for the dyadic decompositions; compare [3, Proposition 4.13].

Proposition 4.1. *Fix n, k as in Theorem 1.1. Given $\eta > 0$, one can choose $\kappa > 0$ sufficiently small and then $a_0 > 0$ arbitrarily small such that, with $K \asymp R^{\kappa}$, the following holds:*

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_R)} \lesssim_{\eta} R^{2\eta}\|f\|_2^2 \max_{\theta:\, R^{-1/2}\text{-}cap} \|f\|^{p_b-2}_{L^2_{\mathrm{avg}}(\theta)}, \qquad A \geqslant R^{a_0}. \tag{4.1}$$

All constants are independent of R and A. For canonical sector decompositions, $\|f\|_2 \lesssim \|f\|_2$.

Proof. **Parameter order.** Choose a stopping exponent $\theta > 0$ with $n\theta < \eta/4$. Choose a time-shortening exponent $\alpha > 0$ small enough that every geometric loss $r^{C_{n,k}\alpha}$ below is bounded by $r^{\eta/4}$. Choose $b_0 > 0$ with $2b_0 < \eta\alpha/4$. Take the wave-packet enlargement exponent and the arbitrary losses in (2.3) and refined decoupling much smaller than b_0 and η. Choose κ so small that the extraction loss $K^{C_{n,k}} \leqslant R^{\eta/4}$ and $K^{2L} \ll R^{\theta}$. Finally take $a_0 > 0$ as small as desired, but fixed and positive. Increase the lower threshold for R so that every polylogarithmic loss is dominated by the positive powers used below.

For $r \in [R^{\theta}, R]$ we prove

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)} \leqslant C_{\eta} R^{\eta} r^{\eta}\|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|^{p_b-2}_{L^2_{\mathrm{avg}}(\theta)}, \qquad A \geqslant r^{a_0}. \tag{4.2}$$

Use a discrete sequence with successive radii r and $r_1 = r^{1-\alpha}$, rounded dyadically. Its length down to R^{θ} is bounded in terms of η alone. For an intermediate requested radius, start the descending sequence at that radius.

Base case. For $r \lesssim R^{\theta}$, the trivial bound is $Cr^n\|f\|_2^{p_b}$. Since $\|f\|_2 \lesssim \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}$ and $r^n \leqslant CR^{\eta/4}$, this is absorbed by (4.2).

Packet selection and broad-mass regularization. At scale r, discard the rapidly decreasing packet tail and select common energy and per-direction-count bins as in (2.9). Lemma 2.4 costs a factor $(\log r)^C$ in the broad parameter and in the norm. Select a union X of $r^{1/2+\epsilon_0}$-cubes on which the broad mass per cube is comparable and which carries the original broad norm up to logarithms.

All subsequent pigeonholing involving the density and the number of active time intervals can be made to preserve both a polylogarithmic fraction of the number of these cubes and the required broad integral. To verify this, if $I_Q \asymp I$ and $I_Q \leqslant L^C \sum_{j=1}^{L} I_{Q,j}$, first choose $j(Q)$ with $I_{Q,j(Q)} \geqslant L^{-C-1} I_Q$, then a value of j occurring on at least $1/L$ of the cubes, and then a dyadic size of $I_{Q,j}$. The relevant sizes have polynomial range in r, after discarding negligible terms.

Common time intervals. Partition the x_n axis into intervals of length $r_1 = r^{1-\alpha}$, aligned with the physical grid. For every tube, define its segments using these *same* time intervals. Select a dyadic λ for which each retained segment J meets

X in volume comparable to $\lambda|T|$. Select a dyadic number β of such segments per retained tube. Necessarily $1 \leqslant \beta \lesssim r^{\alpha}$. Let

$$Y(T) = \bigcup_{J \text{ selected for } T} (J \cap X \cap T)$$

with grid-cell shadings in a fixed tube enlargement. Its density is comparable to $\lambda_0 = \beta\lambda$. On a fixed common time interval the truncated sum is exactly the extension of the subfamily whose interval is active, modulo the already controlled spatial tails. There is no pointwise monotonicity assertion when deleting packets.

The selected broad measure is carried by a set $X_2 \subset X$, with

$$|X_2| \gtrsim (\log r)^{-C}|X|, \tag{4.3}$$

and with comparable selected broad mass on its constituent cubes. The broad parameter after these selections is at least $A/(\log r)^C$.

Case 1: few active time intervals. Suppose $\beta \leqslant r^{b_0}$. Partition each common time interval spatially into cubes of side comparable to r_1. For such a cube B_j, let $\boldsymbol{f}_j$ be the vector of active packets whose tubes meet a fixed enlargement of B_j, retaining sector labels. On B_j the truncated sum agrees with $E\boldsymbol{f}_j$ up to rapidly decreasing errors. A tube meets only $O_n(1)$ spatial cubes in each active interval, since its slope is bounded and its width is $o(r_1)$. Consequently,

$$\sum_j \|\boldsymbol{f}_j\|_2^2 \lesssim r^{b_0+C\epsilon_0}\|\boldsymbol{f}\|_2^2,$$

$$\max_{\theta:\, r_1^{-1/2}\text{-cap}} \|\boldsymbol{f}_j\|_{L^2_{\mathrm{avg}}(\theta)} \lesssim \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}.$$

The average estimate follows from (2.8); the new support buffer absorbs the packet-projection enlargement.

Since

$$\frac{r^{a_0}}{(\log r)^C} \geqslant r^{(1-\alpha)a_0} = r_1^{a_0}$$

for sufficiently large R, the induction hypothesis applies on every B_j. After summation the ratio to the right side of (4.2) is bounded by

$$C(\log r)^C r^{-\eta\alpha+b_0+C\epsilon_0}.$$

The parameter choices make this smaller than $1/2$. This closes the first case.

Case 2: many active time intervals. Suppose $\beta > r^{b_0}$. A ball of radius comparable to r_1 meets only $O_n(1)$ of the common time intervals. Since all selected segments have comparable shading size, its share of any $Y(T)$ is at most C/β. Hence the normalized shading is a two-ends shading with parameters comparable to 2α and b_0 at tube scale $\delta \asymp r^{-1/2}$, with the small enlargement adjustments already budgeted.

Apply (2.2) to this *ancestor* shading before selecting transverse caps. This gives a union of good grid cubes with exceptional volume at most $r^{-c\alpha}|X|$ and the multiplicity bound (3.3). By (4.3) and the uniform broad mass on X_2, the exceptional set removes less than half of the selected broad mass for sufficiently large R. Restrict to the good set. No further two-ends property is required of subsequent restrictions.

Apply Lemma 2.5. Fix a transverse tuple at cost K^C, then one spatial cube in one common time interval at cost at most $r^{C_n\alpha}$. On that cube the active packet families are fixed complete extensions. Pigeonhole their separate positive multiplicities on the $r^{1/2+\epsilon_0}$ grid while retaining the additive functional $\sum_{b\subset X} \min_i \int_b |G_i|^{p_b}$ from

Lemma 2.5; there are only polylogarithmically many joint bins. Choose a bin carrying a polylogarithmic fraction of that functional. On that bin, and not before it is chosen, bound the functional by $(\prod_i \int_X |G_i|^{p_b})^{1/k}$. Cubes where a family has zero multiplicity contribute only controlled tails to the local minimum functional and are removed. The surviving families meet the hypotheses of Subsection 3.2, with the fixed subfamily in the selected time slab used for each G_i. Their average densities are no greater than a constant times the ancestor density, since every packet's new shading is a subset of its ancestor shading. The inherited good-set upper bound is unchanged.

If a multiplicity is small, use the direct low-multiplicity argument there. Otherwise use Lemma 3.1 and (3.4). Combining all factors yields

$$\|E\boldsymbol{f}\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)} \lesssim_\eta K^{C_{n,k}} r^{C_{n,k}\alpha+o_{\epsilon_0}(1)+\epsilon'}(\log r)^C \|\boldsymbol{f}\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{p_b-2}_{L^2_{\mathrm{avg}}(\theta)},$$

where the arbitrary input loss $\epsilon' > 0$ was chosen sufficiently small. The parameter choices absorb this into $C_\eta R^\eta r^\eta \|\boldsymbol{f}\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{p_b-2}_{L^2_{\mathrm{avg}}(\theta)}$. The constant C_η can be enlarged once to cover the base scales. This completes the induction and proves (4.1). □

4.2. **From mixed broad estimates to linear restriction.** Fix

$$\max\{p_b, p_N\} < p < \frac{2(k-1)}{k-2},$$

where the upper endpoint is interpreted as infinity when $k = 2$. By Proposition 4.1, the bound $\max_{\theta:\, R^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)} \leqslant \|f\|_\infty$, and $\|Ef_\tau\|_\infty \lesssim \|f\|_\infty$, we have

$$\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} \lesssim_{\eta,p} R^{2\eta}\|f\|_2^2\|f\|_\infty^{p-2}, \qquad K \asymp R^\kappa, \quad A \geqslant R^{a_0}. \tag{4.4}$$

We use this estimate directly in the narrow induction, with target

$$\int_{B_R} |Ef|^p \leqslant C_\epsilon R^\epsilon \|f\|_2^2 \|f\|_\infty^{p-2}. \tag{4.5}$$

The broad–narrow decomposition and [6, Lemma 9.3] give, for every $\delta_N > 0$,

$$\begin{aligned}\int_{B_R} |Ef|^p \lesssim\ & K^{C_{n,p}} \|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} \\ & + C_{\delta_N} A^{p-1} K^{\delta_N + (k-2)(p/2-1)} \sum_\tau \int_{B_{2R}} |Ef_\tau|^p.\end{aligned} \tag{4.6}$$

Here each narrow cap is assigned to one exceptional plane; Hölder over the resulting disjoint groups accounts for A^{p-1}.

For a cap τ of radius K^{-1}, let $g_\tau(\xi) = f_\tau(\xi_\tau + K^{-1}\xi)$. Then

$$\|g_\tau\|_2^2 = K^{n-1}\|f_\tau\|_2^2, \qquad \|g_\tau\|_\infty = \|f_\tau\|_\infty.$$

Parabolic rescaling maps B_{2R} into a ball of radius CR/K. Applying the induction hypothesis there yields

$$\int_{B_{2R}} |Ef_\tau|^p \lesssim C_\epsilon (R/K)^\epsilon K^{2n-(n-1)p} \|f_\tau\|_2^2 \|f_\tau\|_\infty^{p-2}.$$

Since the frequency cutoffs have bounded overlap,

$$\sum_\tau \|f_\tau\|_2^2 \|f_\tau\|_\infty^{p-2} \lesssim \|f\|_2^2 \|f\|_\infty^{p-2}.$$

Consequently, setting

$$\nu_p = \frac{2n-k}{2}p - (2n-k+2) > 0,$$

the narrow contribution in (4.6) is bounded by

$$CC_\epsilon R^\epsilon A^{p-1} K^{-\nu_p+\delta_N-\epsilon} \|f\|_2^2 \|f\|_\infty^{p-2}.$$

Choose the parameters in Proposition 4.1 so that

$$2\eta + \kappa C_{n,p} < \frac{\epsilon}{2}, \qquad \delta_N < \frac{\nu_p}{4}, \qquad \frac{a_0(p-1)}{\kappa} < \frac{\nu_p}{4},$$

and take $A = \lceil R^{a_0} \rceil$. This is possible since a_0 may be chosen arbitrarily small after κ is fixed. The factor $A^{p-1}K^{-\nu_p+\delta_N-\epsilon}$ is then $O(K^{-\nu_p/2})$, so the narrow term is absorbed into (4.5) for sufficiently large R. Meanwhile, (4.4) bounds the broad term by

$$C_{\eta,p} R^{\epsilon/2} \|f\|_2^2 \|f\|_\infty^{p-2}.$$

Taking C_ϵ sufficiently large to cover this term and the bounded base scales closes the induction. Finally, restricted-type real interpolation and a standard ϵ-removal argument (see [11]) complete the proof.

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